

Decision-Point Shared Control for Self-Calibrating Motor-Imagery BCI Navigation of a Mobile Robot

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Abstract—Motor-imagery (MI) brain-computer interfaces (BCIs) are attractive for hands-free robot control but are limited by noisy, non-stationary electroencephalography (EEG): decoders trained offline degrade across sessions, and most systems require per-user training data and continuous decoding. We present a *decision-point* shared-control framework in which the BCI is queried only at navigation junctions: a mobile robot drives autonomously along corridors and, on reaching a wall, halts and waits for a single left/right MI command that selects the turn. The decoder is trained once and *frozen*; at the start of each session an unsupervised, label-free calibration recomputes only the covariance alignment reference, and an adaptive *boundary-recentering* rule continuously tracks the user’s neutral operating point so a fixed model stays calibrated online without retraining. We evaluate two decoders—Euclidean-Alignment Filter-Bank CSP with shrinkage LDA, and a Riemannian tangent-space pipeline with logistic regression—crossed with personalized (per-user) and generalized (multi-user, pooled) training, in a 2×2 in-the-loop study on four users driving a first-person robot maze simulator. Across 16 trials the users completed every maze (16/16), with 92.0 % first-attempt *corner* accuracy and a median command latency of 0.20 s. A single generalized decoder performed comparably to per-user decoders at the task level, and because decision-point shared control renders command errors recoverable, imperfect per-command decoding still yielded full task completion.

Index Terms—Brain-computer interface, motor imagery, shared control, mobile robot navigation, EEG, transfer learning, human-robot interaction.

I. INTRODUCTION

BRAIN-computer interfaces (BCIs) based on motor imagery (MI) let a user issue commands by imagining movements, offering a control channel for people with severe motor impairment [1], [2]. Translating MI-BCIs from offline benchmarks to live robot control, however, remains difficult for three reasons. First, EEG is noisy and *non-stationary*: a decoder trained in one session loses accuracy in the next as electrode montage, impedance, and user state drift [3], [4]. Second, most pipelines require *per-user* training data, adding an enrollment burden. Third, *continuous* decoding forces the user to sustain imagery throughout operation and exposes every window to misclassification, which for a moving robot can be unsafe.

We address these issues through the interaction design rather than through the classifier alone. Our premise is that navigation only needs *discrete* choices at *decision points* (corridor junctions); between junctions the robot can drive autonomously.

This work was conducted under an approved institutional review board (IRB) protocol.

This reduces the BCI’s role to one binary command per junction and lets the system reject or defer during ambiguous intervals. We combine this with an *unsupervised, label-free* online adaptation scheme that keeps a frozen decoder calibrated to the current session without retraining.

Prior work in our line established accurate *offline* MI classification and quantified the across-session accuracy loss—the “laboratory-to-practice” gap [5], [3]. This paper closes that gap in a live navigation loop and evaluates it at the *task* level. Our contributions are:

- 1) A **decision-point shared-control** navigation framework that restricts BCI involvement to junction turn-choices (about one command per corner, 0.20 s median latency), with autonomous heading-hold and a range-based stop between junctions (Sec. IV-D).
- 2) An **unsupervised, label-free session-adaptation** scheme—alignment-reference recomputation plus adaptive decision-boundary recentering—that keeps a *frozen* decoder calibrated online with no labels and no retraining (Sec. IV-B).
- 3) A 2×2 **in-the-loop study** (two decoders $\times$ personalized/generalized training) evaluating both decoder families and the effect of pooling users’ data, at the task level (Sec. VII).
- 4) A **task-level evaluation** distinguishing per-corner from per-command accuracy and showing that recoverable errors under shared control convert $\sim 92\%$ per-corner decoding into 100 % task completion.

We evaluate in a kinematic robot-maze *simulator*; the scope and its limits are stated explicitly in Secs. V and IX.

II. RELATED WORK

MI decoding. Common Spatial Patterns (CSP) [6], [7] and its filter-bank extension (FBCSP) [8], [9] with (shrinkage-regularized) LDA [10], [11] remain strong, data-efficient baselines. Covariance-manifold methods classify EEG spatial covariances directly on the symmetric positive-definite manifold via Riemannian distances and tangent-space projection [12], [13], [14], [15]. Deep networks (e.g., EEGNet, deep/shallow ConvNets, transformers) are the modern alternative [16], [17], [18], [19], [20] but are data-hungry and heavier to deploy; we adopt lightweight decoders suitable for embedded, real-time use. EEG intent recognition for human-robot interaction on comparable acquisition hardware has been studied offline with FB-CSP and linear classifiers [21], [22].

Cross-session/subject transfer. Alignment methods reduce distribution shift: Euclidean Alignment (EA) recentres raw

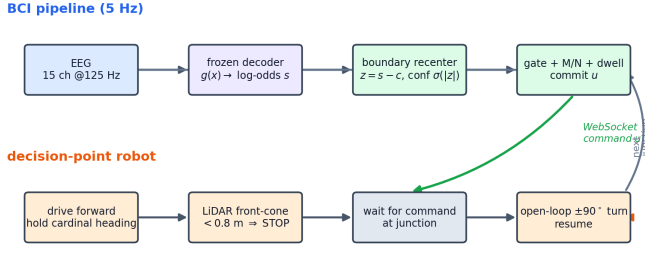


Fig. 1. Decision-point shared-control loop. The frozen BCI pipeline (top) emits committed left/right commands at 5 Hz after recentering and gating; the robot (bottom) drives autonomously, stops at a wall, executes the commanded turn, and proceeds to the next junction.

trials by their covariance mean [23], while Riemannian alignment and Procrustes analysis transport covariance distributions on the manifold [24], [25]; subject-independent deep models pursue the same goal [26], [4]. These correct the covariance statistics but not necessarily the classifier’s operating point, which we address online.

BCI-controlled robots. MI and hybrid BCIs have driven wheelchairs and mobile robots [27], [28], [29], [30], [31], robotic arms [32], [33], wheelchairs with automated navigation [34], and telepresence platforms [1], increasingly with *shared control* that blends user intent with autonomy [30], [35], [2]. Asynchronous/self-paced control and command rejection avoid forcing a decision every window [36], [29], and error-related potentials have been used to detect mistakes [37], [38]. Our decision-point scheme is in this tradition but pairs discrete junction commands with an unsupervised, label-free recalibration of a frozen decoder and reports task-level rather than only classification metrics.

III. SYSTEM OVERVIEW AND PROBLEM FORMULATION

The system (Fig. 1) has three parts: EEG acquisition and decoding, an unsupervised online adaptation stage, and a decision-point robot controller communicating over a Web-Socket. EEG is acquired from an OpenBCI Cyton+Daisy stack (15 usable channels at the 10–20 sensorimotor sites, sampled at $f_s = 125$ Hz). Every stride ($\Delta t = 0.2$ s) the most recent 2.0 s window is decoded, yielding a 5 Hz stream of tentative left/right decisions that are smoothed and, when confident and stable, committed as a command.

Problem. Let $x \in \mathbb{R}^{C \times T}$ be a band-limited EEG window (C channels, T samples). A frozen decoder produces a log-odds score $s = g(x) \in \mathbb{R}$ toward the “right” class. Because the score distribution shifts across sessions and users, a fixed threshold on s is unreliable. We seek an unsupervised, label-free mapping from s to a committed command $u \in \{\text{left}, \text{right}, \emptyset\}$ that (i) tracks each session’s operating point without retraining g , and (ii) abstains ($\emptyset$) when evidence is weak, so that the robot acts only on reliable commands at decision points.

IV. PROPOSED METHOD

A. Decoders

Both decoders share the interface g : window $\rightarrow$ log-odds, and a filter bank of four bands $\mathcal{B} = \{(8, 12), (11, 15), (14, 20), (20, 30)\}$ Hz realised with causal minimum-phase FIR filters, so decoding is real-time.

(A) *EA + FB-CSP.* Per band b and trial i , the spatial covariance is $C_i = \frac{1}{T} X_i X_i^\top$ and the Euclidean-Alignment reference is the mean $R = \frac{1}{N} \sum_i C_i$ [23]. With $R = V \Lambda V^\top$, whitening each trial by $R^{-1/2} = V \Lambda^{-1/2} V^\top$ makes the aligned mean covariance identity:

$$\tilde{X}_i = R^{-1/2} X_i, \quad \frac{1}{N} \sum_i \tilde{C}_i = I. \quad (1)$$

CSP then extracts two components per band; the feature is their log-variance, and a shrinkage-LDA [11] produces the log-odds $s = w^\top f + b_0$.

(B) *Riemannian tangent-space.* Per band, a Ledoit–Wolf covariance C_i is recentred by the Riemannian (Fréchet) mean $\bar{C} = \arg \min_M \sum_i \delta_R(M, C_i)^2$, where $\delta_R(A, B) = \|\log(A^{-1/2} B A^{-1/2})\|_F$ [12], [24]; the recentred matrices are mapped to the tangent space at the identity, $v_i = \text{upper}(\log(\bar{C}^{-1/2} C_i \bar{C}^{-1/2}))$ [13], and a logistic regression gives $s = w^\top v + b_0$.

At deployment the model weights are *frozen*. A 45 s calibration block recomputes only the alignment reference (R or $\bar{C}$) from the live session, absorbing the covariance shift without labels.

B. Unsupervised boundary recentering and gating

Alignment corrects covariance statistics but not where the classifier’s boundary sits, so a session’s log-odds can be offset (a neutral/rest state read as a confident class). We track the neutral operating point online. Seeding from the calibration median $c_0 = \text{median}_j s_j$, the recentred margin, committed side, and confidence are

$$z_t = s_t - c_t, \quad \hat{y}_t = \begin{cases} \text{right}, & z_t > 0 \\ \text{left}, & z_t \leq 0 \end{cases}, \quad (2)$$

$$\text{conf}_t = \sigma(|z_t|) = \frac{1}{1 + e^{-|z_t|}}. \quad (3)$$

The neutral is updated only from *rest-like* windows (small margin), so sustained imagery never drags the boundary, and it is clamped near its seed:

$$c_{t+1} = \text{clip}(c_t + \alpha(s_t - c_t), c_0 - \Delta, c_0 + \Delta) \text{ if } |z_t| < \tau. \quad (4)$$

A window is eligible to commit only if $\text{conf}_t \geq \theta$; because confidence is measured from the recentred margin, this gate is also a *dead-zone* around neutral, so idle/rest does not commit (Fig. 2). We use $\alpha = 0.01$, $\Delta = 2.0$, $\theta = 0.85$ (Table I).

C. Temporal smoothing and commitment

Gated per-window decisions feed an M -of- N majority vote ($N=5$, $M=4$) followed by a dwell requirement ($D=3$ consecutive consensus windows) before a command is committed and sent to the robot. This tolerates isolated misclassifications while keeping commitment latency low.

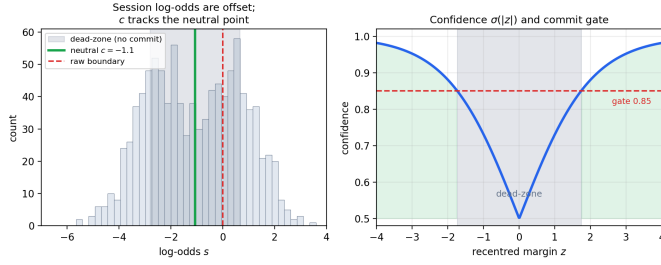


Fig. 2. Unsupervised boundary recentering. Left: a session’s log-odds are offset, so the adaptive neutral c (green) sits away from the raw boundary. Right: confidence $\sigma(|z|)$ with the commit gate; the gate forms a dead-zone around neutral so weak/rest evidence abstains.

D. Decision-point controller

Between junctions the robot drives forward at constant speed while a proportional law holds the nearest cardinal heading. A simulated planar LiDAR provides a front-cone ($\pm 15^\circ$) minimum range; when it falls below 0.8 m the robot stops and enters a waiting state at the decision point. It remains stopped until a committed BCI command arrives, then executes an open-loop $\pm 90^\circ$ turn about the snapped cardinal heading (which prevents heading drift from accumulating) and resumes. If the turn is wrong the robot faces a wall, re-stops, and waits again—so an error is locally *recoverable* at the cost of additional commands, never a crash or an off-course excursion.

V. EXPERIMENTAL METHODOLOGY

Simulation scope. Experiments use a first-person robot-maze simulator with a TIAGo/PMB2-shaped robot. The simulator integrates a kinematic unicycle pose at a fixed timestep and models collisions geometrically; it is *not* a physics/dynamics engine, and autonomy is limited to heading-hold and range-based stopping. It provides a controlled, repeatable test of the BCI-shared-control loop; physical-robot validation is future work (Sec. IX).

Data and decoders. Offline MI data were collected from eight participants under an approved IRB protocol, each contributing several sessions of cued left/right-hand imagery (one further participant was excluded for signal-quality reasons). Personalized decoders are trained on the driving user’s own data; generalized decoders are trained on the pooled data of all eight participants *including* the driving user (i.e., a single shared multi-user model, not a subject-held-out model).

Task and design. Each trial navigates a procedurally generated 40-corner maze. We use a 2×2 design—decoder {EA+FB-CSP, Riemannian-TS+LR} $\times$ training {personalized, generalized}—with one trial per cell per user and maze seeds counterbalanced across users. Four users each completed the four conditions, for 16 live trials.

VI. EVALUATION METRICS

We separate command-level from task-level performance:

- **Corner accuracy** (first-attempt): fraction of the 40 corners at which the *first* committed command was correct—the per-intent decoding reliability.

TABLE I
SYSTEM AND DECODER PARAMETERS.

Symbol / setting	Value	Meaning
f_s	125 Hz	EEG sampling rate
C	15	usable EEG channels
window / stride	2.0 s / 0.2 s	decode window; 5 Hz decisions
B	8–30 Hz (4 bands)	filter-bank passbands
CSP comps. / band	2	EA+FB-CSP feature dimension
calibration	45 s	unsupervised reference recompute
θ	0.85	commit / dead-zone gate on conf.
α, Δ	0.01, 2.0	recenter rate; clamp (log-odds)
N, M, D	5, 4, 3	vote window, majority, dwell

TABLE II
OFFLINE LEFT/RIGHT MI DECODING ACCURACY (8 PARTICIPANTS;
MEAN $\pm$ STD OVER 5-FOLD CV).

Decoder	Personalized (within-subj.)	Generalized (pooled)
EA+FB-CSP	91.1 $\pm$ 6.3 %	84.8 $\pm$ 0.6 %
Riemann-TS+LR	89.5 $\pm$ 8.9 %	77.6 $\pm$ 3.0 %

- **Turn accuracy** (per-command): fraction of *all* committed commands that were correct, including recovery commands after an error.
- **Completion**: fraction of mazes fully navigated to the goal.
- **Commands per corner**: total commands / 40 (recovery overhead; 1.0 is ideal).
- **Command latency**: wait time from stopping at a junction to a committed command.

VII. RESULTS

Offline decoding. Both decoders separated left/right imagery well (Table II). Personalized within-subject cross-validation reached 91.1 $\pm$ 6.3 % (EA+FB-CSP) and 89.5 $\pm$ 8.9 % (Riemannian-TS+LR); pooled generalized models scored 84.8 $\pm$ 0.6 % and 77.6 $\pm$ 3.0 %, respectively. The generalized EA+FB-CSP model retained most of the personalized accuracy, whereas the Riemannian pipeline lost more when pooled—a pattern that recurs in the loop.

In-the-loop navigation. All 16 trials were completed (16/16). Pooled over trials, first-attempt **corner accuracy was 92.0 %** (589/640), while per-command **turn accuracy was 81.3 %** (665/818); the gap reflects recovery commands and shows why the corner metric is the meaningful measure of decoding reliability. Recovery overhead was 1.28 commands/corner. Median **command latency was 0.20 s** (mean 0.52 s, 90th percentile 1.30 s), with 81 % of commands issued within 0.5 s (Fig. 4).

Table III reports the 2×2 conditions. Averaged over training, the two decoders were comparable (EA+FB-CSP 92.8 % vs. Riemannian-TS+LR 91.2 % corner accuracy). Averaged over decoder, personalized training gave 94.1 % and generalized 90.0 % corner accuracy: a single *generalized* model shared across users performed close to per-user models, and the EA+FB-CSP generalized decoder (93.1 %) essentially matched its personalized counterpart (92.5 %). The Riemannian decoder was the most sensitive to this factor (95.6 % personalized vs.

TABLE III
IN-THE-LOOP TASK-LEVEL RESULTS BY CONDITION (4 USERS, 4 TRIALS EACH).

Condition	Corner acc.	Turn acc.	Cmd/corner
EA+FB-CSP, personalized	92.5 %	77.5 %	1.39
Riemann-TS+LR, personalized	95.6 %	91.0 %	1.11
EA+FB-CSP, generalized	93.1 %	85.8 %	1.19
Riemann-TS+LR, generalized	86.9 %	73.7 %	1.43
Pooled	92.0 %	81.3 %	1.28

TABLE IV
PER-USER TASK-LEVEL RESULTS (POOLED OVER THE FOUR CONDITIONS).

User	Corner acc.	Turn acc.	Med. wait	Med. time
U1	87.5 %	72.2 %	0.20 s	389 s
U2	96.2 %	92.5 %	0.15 s	313 s
U3	88.8 %	76.1 %	0.20 s	357 s
U4	95.6 %	88.3 %	0.15 s	339 s

86.9 % generalized). Per-user corner accuracy ranged from 87.5 % to 96.2 % (Table IV). Fig. 5 shows example trajectories, including one with an error-recovery loop.

VIII. DISCUSSION

Three findings stand out. First, *decision-point shared control makes the BCI's job small and its errors cheap*: one command per junction, a sub-second median latency, and local recovery that turned $\sim 92\%$ per-corner decoding into full task completion. Second, *a single generalized model is competitive*: pooling users' data (including the driving user) matched per-user models within a few points for EA+FB-CSP, suggesting that maintaining one shared decoder rather than one per user is a viable deployment simplification—though, because the pool includes the user, this is a shared-model result, not a zero-shot subject-independent one. Third, *the online neutral point genuinely drifts* across sessions (Fig. 6), which a fixed threshold cannot follow; the unsupervised recentering tracks it without labels or retraining, consistent with the frozen-decoder design. The two decoder families were comparable on average, with the Riemannian pipeline more sensitive to whether training was personalized.

IX. LIMITATIONS

The evaluation is in a *kinematic simulator* with limited autonomy (heading-hold and range-based stopping; no dynamics, wall-following, or planning), so absolute times and the control law would differ on hardware; physical-robot validation is the essential next step. The in-the-loop study has four users and one trial per condition, which supports descriptive comparison but not strong statistical inference; a larger cohort with repeated measures is needed. The generalized condition includes the driving user's data, so we make no subject-independent claim. Finally, the waiting state is unbounded (no timeout), and we did not isolate the recentering component with an in-loop ablation—both are targeted in ongoing work.

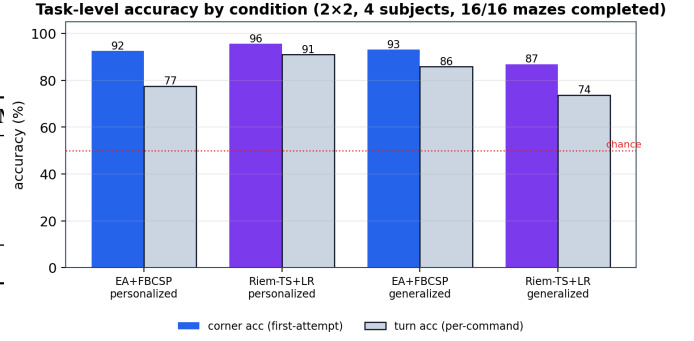


Fig. 3. Task-level accuracy by condition. Corner accuracy (first-attempt, coloured) vastly exceeds chance; per-command turn accuracy (grey) is lower because it counts recovery commands. All 16 mazes were completed.

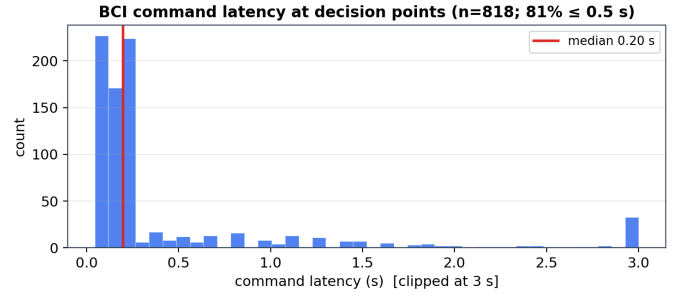


Fig. 4. Distribution of BCI command latency at decision points ($n=818$ commands); median 0.20 s, 81 % within 0.5 s.

X. CONCLUSION

We presented a decision-point shared-control framework that lets a frozen, self-calibrating MI-BCI drive robot navigation, keeping the decoder aligned to each session with unsupervised, label-free recentering. In a 2×2 in-the-loop study, users completed every maze with 92 % per-corner accuracy and 0.20 s median command latency, and a single generalized decoder was competitive with per-user models. The results indicate that careful interaction design—asking the BCI only at decision points and making errors recoverable—can yield reliable task-level performance from imperfect EEG decoding. Physical-robot validation, a larger cohort, and an in-loop ablation of the recentering component are the priorities for future work.

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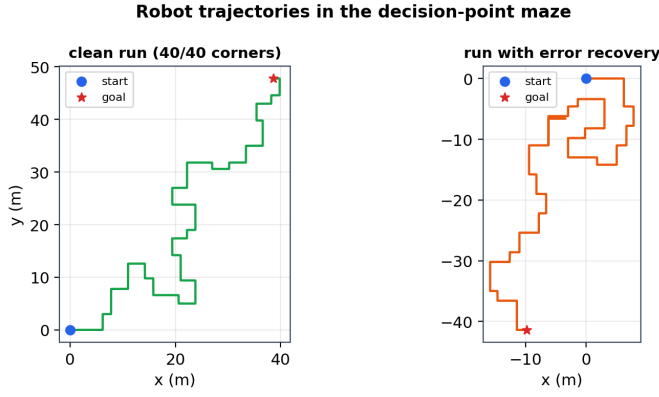


Fig. 5. Example robot trajectories: a clean run (left) and a run with an error-recovery loop (right); both reach the goal.

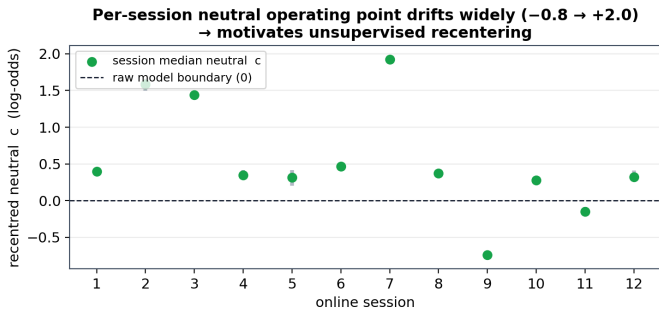


Fig. 6. The per-session neutral operating point c varies widely across online sessions (-0.8 to $+2.0$ log-odds), motivating unsupervised recentering of a frozen decoder.

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